

# A Naked Mole Rat's Anti-Aging Shield Has Been Successfully Transferred to Mice--and It Worked

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Naked mole rats are biological rule-breakers. These mouse-sized rodents can live up to 41 years nearly ten times longer than their furry cousins and they almost never develop cancer, even when researchers try to induce it. For decades, scientists have been trying to unpack their secrets. Now, a team at the University of Rochester has shown that one of those secrets is portable. In a landmark study published in *Nature*, researchers transferred a gene responsible for **high molecular weight hyaluronic acid** (HMW-HA) from naked mole rats into mice. The result: the genetically modified mice lived healthier, with fewer spontaneous and chemically induced tumors, reduced **inflammation** across multiple tissues, and a 4.4% increase in **median lifespan**. The key player is a version of the **hyaluronan synthase 2** gene, which drives production of HMW-HA a long, sugar-like molecule that forms a gooey matrix around cells. Naked mole rats have roughly ten times more HMW-HA than humans or mice. Earlier work showed that stripping this molecule from mole rat cells made them vulnerable to tumors, hinting at its protective role. When the Rochester team engineered mice to express the mole rat version of the gene, the animals produced more HMW-HA and showed broad health improvements. **Inflammation**, a hallmark of aging, was dialed down. Gut health improved. And the mice simply lived longer. "Our study provides a **proof of principle** that unique longevity mechanisms that evolved in **long-lived mammalian species** can be exported to improve the lifespans of other mammals," said Vera Gorbunova, the study's senior author. The lifespan gain was modest, but the implications are enormous. It suggests that nature's longest-lived creatures harbor molecular tools that might one day be adapted to delay aging in humans. The team's next goal? To translate this benefit to people. It's a reminder that sometimes the most profound breakthroughs come from the strangest places like a bald, buck-toothed rodent living underground.

## Word Treasure

**high molecular weight hyaluronic acid** -- A very long-chain sugar-like molecule that surrounds cells and helps protect them; abundant in naked mole rats.

**hyaluronan synthase 2** -- A gene that instructs the body to produce an enzyme building hyaluronic acid, especially the high molecular weight form.

**inflammation** -- The body's protective response to injury or harmful agents, often linked to aging when it becomes chronic.

**median lifespan** -- The age at which half of a study group is still alive - a statistical measure of typical life length.

**proof of principle** -- An experiment that demonstrates a concept can work, even if it's not yet a practical application.

**long-lived mammalian species** -- Mammals that naturally live much longer than similar-sized relatives, like naked mole rats compared to

mice.

## Background you need

Naked mole rats (*Heterocephalus glaber*) are native to East Africa and live in large underground colonies with a single breeding queen - a social structure almost unique among mammals, resembling ants or bees.

Scientists at the University of Rochester, including senior author Vera Gorbunova (a leading biologist in aging research), have studied these animals for over a decade. An earlier 2013 study first showed that their unusually dense hyaluronic acid was key to cancer resistance.

Hyaluronic acid is already used in some skincare products and medical treatments, but the naked-mole-rat version is much larger and seems to create a physical barrier around cells that prevents tumor growth and reduces oxidative damage.

The study published in *Nature* - one of the most prestigious scientific journals, founded in 1869 - represents a major step in gerontology (the science of aging). Many researchers hope that by understanding natural longevity mechanisms, we can help people live healthier lives even if maximum lifespan doesn't change dramatically.

## Break it down

**HOOK:** Naked mole rats are biological rule-breakers - up to 41 years lifespan, almost no cancer.

**STUDY ANNOUNCEMENT:** University of Rochester team transfers a key gene to mice.

+ RESULTS: Healthier mice, fewer tumors, less inflammation, 4.4% longer median lifespan.

**MECHANISM EXPLAINED:** The hyaluronan synthase 2 gene produces HMW-HA.

+ COMPARISON: Naked mole rats have ~10 times more HMW-HA than humans or mice.

+ PRIOR EVIDENCE: Removing HMW-HA made mole rat cells vulnerable to tumors.

**QUOTE FROM EXPERT:** Vera Gorbunova calls it a 'proof of principle' that longevity mechanisms can be exported across species.

**FUTURE DIRECTION:** Translating the benefit to humans.

+ BROADER IMPLICATION: Strange animals may unlock molecular tools to delay aging.

## Why it matters

Aging and cancer affect everyone, and this study shows that solutions might come from studying nature's most unusual creatures. For a middle schooler, it's a reminder that biological diversity is not just interesting - it could one day lead directly to medicines that help their grandparents or even themselves stay healthier longer.

## Test what you remember

1. How much longer was the median lifespan of the genetically modified mice, according to the study?
  - (A) 1.2%
  - (B) 4.4%
  - (C) 10%
  - (D) 15%
2. What molecule did the transferred gene increase in the mice?
  - (A) Collagen
  - (B) High molecular weight hyaluronic acid (HMW-HA)
  - (C) Hemoglobin
  - (D) Dopamine
3. What is unusual about naked mole rats compared to other rodents?
  - (A) They can fly short distances
  - (B) They can live up to 41 years and rarely get cancer
  - (C) They hibernate for decades
  - (D) They glow in the dark
4. Which university is the senior author Vera Gorbunova from?
  - (A) Stanford University
  - (B) University of Rochester
  - (C) Harvard University
  - (D) University of Cambridge
5. What is the team's next goal after this mouse study?
  - (A) To sequence the entire naked mole rat genome
  - (B) To test the therapy in dogs
  - (C) To translate the benefit to humans
  - (D) To breed naked mole rats with mice
6. What happened when scientists previously removed HMW-HA from naked mole rat cells?
  - (A) The cells lived even longer
  - (B) The cells became vulnerable to tumors
  - (C) The cells stopped dividing completely
  - (D) The cells turned into cancer stem cells

Answer key (try the quiz first!): 1: B 2: B 3: B 4: B 5: C 6: B

